

QUEUE

Q10)

Question:

DS	Session	Queue	Question Information	Level 1	Challenge 60
Question description					
Umesh is an DS expert training youngsters struggling in DS to make them better.					
Umesh usually gives interesting problems to the youngsters to make them love the DS.					
One such day Umesh provided to the youngsters to solve the task such that, Reverse a Queue, Queue data structures work on the FIFO architecture so the element that has entered first in the list will go out from the list first.					
Youngsters were lacking the idea to solve the problem.					
Being an exciting youngster can you solve it?					
Function Description					
• INITIALIZE FRONT AND REAR = -1 • CALL ENQUEUE FUNCTION WHILE(I<GIVEN SIZE OF QUEUE) • CALL REVERSE • MAKE A FOR LOOP(I=FRONT,J=REAR;I<J;I++;J++) • SWAP(FRONT,REAR)					
Constraints:					
0<size <100					
0<data<1000					
Input format:					
First line indicates the size of the queue					
Second line indicates the elements of the queue.					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int n, q[100], front = 0, rear = -1;
5
6      scanf("%d", &n);
7
8      for (int i = 0; i < n; i++) {
9          scanf("%d", &q[++rear]);
10     }
11
12     printf("Queue:");
13     for (int i = front; i <= rear; i++)
14         printf(" %d", q[i]);
15
16     printf("\nReversed Queue:");
17     for (int i = rear; i >= front; i--)
18         printf(" %d", q[i]);
19
20     return 0;
21 }
```

OUTPUT:

TEST CASE 1:

```
5
5 6 3 1 2
Queue: 5 6 3 1 2
Reversed Queue: 2 1 3 6 5
```

TEST CASE 2:

```
8
1 8 6 4 9 5 3 5
Queue: 1 8 6 4 9 5 3 5
Reversed Queue: 5 3 5 9 4 6 8 1
```